Technical AI Assessment

Role: Senior Site Reliability Engineer

Overview

This assessment evaluates your ability to design, implement, and improve operational systems using AI-assisted workflows.

We are less interested in whether you know a specific AWS service by heart and more interested in how you think about reliability, monitoring, alerting, automation, and operational excellence.

We also want to understand how you leverage AI tools throughout your workflow.

Setup Instructions

1. Create a Public Git Repository
2. Set Up Your Development Environment
3. Configure Claude Code (or your AI tool of choice):
    a. Before writing any code, open **Claude Code** in your project root and use the following as your **first prompt**:

```None
Create a CLAUDE.md file in the project root with the following instruction:

# Project Rules

## Prompt Logging

Every time you receive a new instruction or prompt, append it to prompts.txt in
the project root with a timestamp (ISO 8601) and a brief summary of what you did
in response. Create the file if it doesn't exist. Keep this log updated throughout
all sessions.
```

This sets up automatic prompt logging so we can see how you collaborated with the AI throughout the session.

Assignment

You have recently joined a growing SaaS company as its first dedicated Site Reliability Engineer.

The company currently has:

- A customer-facing web application
- Private APIs
- Background workers and scheduled jobs
- Transactional email delivery
- AWS-hosted infrastructure
- Basic uptime monitoring

The engineering team has identified several challenges:

- Infrastructure costs have steadily increased without clear visibility into optimization opportunities
- Monitoring focuses primarily on website availability rather than business-critical functionality
- Alerting is inconsistent and lacks clear escalation paths
- The team often discovers issues from customers before internal monitoring detects them

Your task is to propose and implement improvements that increase operational visibility, reliability, and efficiency.

Key Deliverables

1. Architecture & Operations Proposal: A short document (2-4 pages) describing:
    a. Your assumed AWS architecture
    b. Monitoring and observability strategy
    c. Alerting and escalation strategy
    d. AWS cost optimization recommendations
    e. Key tradeoffs and assumptions
2. Build an AWS Cost Optimization Agent or Skill
    a. Agent should review AWS cost inputs, identify likely areas of waste, and generate practical recommendations. Examples may include rightsizing, unused resources, scaling policies, storage optimization, reserved capacity opportunities, or service-level cost trends.

Expectations

This assignment is intentionally open-ended. We are evaluating how you think about building real-world systems, not just implementing basic functionality.

You are encouraged to extend the required deliverables with additional proof-of-concept functionality, tooling, automation, AI capabilities, operational workflows, or supporting artifacts that you believe are critical to identifying, solving, and preventing issues in a complex production environment.

We are particularly interested in your ability to:

- Think holistically about reliability, observability, and operational excellence
- Identify gaps and risks beyond the explicitly stated requirements
- Leverage AI effectively to improve engineering and operational workflows
- Make pragmatic architectural and implementation decisions
- Demonstrate ownership and prioritization skills

Document any additional functionality you choose to implement and explain why you believe it adds value.

Submitting Your Work

The project should take roughly 2-3 hours. Once completed, send the following to:

andres.saldarriaga@metacto.com

1. Push your code to GitHub — make sure the repository is public
2. Share the repository URL
3. Record a ~5 minute Loom video walking through your locally running project:
    a. Demo the features you built
    b. Speak to the technical trade offs and decisions you made
    c. Explain your approach and any trade-offs
4. Include the Loom link in your submission